

Weekly Diary – Week 3

Internship Organization: Vikaspedia

Duration: 01/02/2026 – 07/02/2026

Objective of the Week

The main objective of this week was to identify and resolve the issue related to the Hindi language in the Text-to-Speech (TTS) system used in the project. During earlier testing of the multilingual TTS highlighting application, it was observed that the Hindi voice was not functioning properly in the browser. Because of this issue, the application was unable to read Hindi text aloud even though other languages were working correctly.

Therefore, the primary goal for this week was to carefully analyze the problem, understand the reason why the Hindi voice was not available, and enable Hindi speech synthesis by installing and configuring the required language components at the system level. In addition to solving the issue, the task also involved testing the solution across different web browsers and documenting the complete troubleshooting process. This documentation would help in understanding the problem better and also serve as a reference for solving similar issues in the future.

Date – 1 February

On the first day, I reviewed the Text-to-Speech application that had been developed earlier in January. This application was designed to convert written text into spoken audio while highlighting the words in real time as they were being spoken. The system supported multiple languages and was built using modern web technologies.

While performing detailed testing of the multilingual features, I noticed that the Hindi language was not functioning as expected. When Hindi text was selected and speech playback was started, the application did not produce any audio output. However, the highlighting feature continued to work correctly. This observation suggested that the problem was specifically related to the speech synthesis voice rather than the highlighting mechanism or the core functionality of the application. I documented the issue and began analyzing the possible reasons behind it.

Date – 2 February

On the second day, I started investigating how browsers provide speech synthesis voices for Text-to-Speech applications. For this purpose, I studied the documentation of the Web Speech API, which is responsible for enabling speech synthesis functionality in web applications.

To understand the issue more clearly, I used the `speechSynthesis.getVoices()` function to retrieve the list of voices available in the browser. This function returns all the speech synthesis voices that the browser can use for generating audio. After printing the list of voices in the console, I carefully examined the available languages. During this inspection, I discovered that there was no Hindi voice present in the list.

This finding indicated that the browser did not have access to a Hindi speech voice on the system. Therefore, the problem was likely related to missing language support in the operating system rather than an error in the application code itself.

Date – 3 February

On the third day, I focused on resolving the missing language support problem. I opened the operating system language settings and searched for available Hindi language options. After exploring the system settings, I installed the Hindi language pack along with the necessary speech components required for Text-to-Speech functionality.

The installation process involved several steps such as adding Hindi as an additional system language, downloading the speech recognition and speech synthesis components, and configuring the language preferences within the system settings. These steps were necessary because many browsers rely on the operating system's installed speech voices to provide speech synthesis capabilities.

Date – 4 February

After installing the required language components, I restarted the system to ensure that the changes were applied properly. Once the system was restarted, I opened the application again and repeated the testing process.

This time, when I executed the `speechSynthesis.getVoices()` function in the browser console, I noticed that the Hindi voice was now included in the list of available voices. This confirmed that the installation of the Hindi language pack had successfully enabled Hindi speech synthesis support on the system.

I then selected the Hindi voice in the application settings and tested the speech playback again. The application was finally able to generate audio output in Hindi, indicating that the issue had been successfully resolved.

Date – 5 February

On the fifth day, I conducted additional testing to verify that the Hindi speech synthesis was working correctly. I used multiple Hindi sample texts to check the pronunciation, clarity, and overall speech output. This testing helped ensure that the Text-to-Speech engine could properly read and pronounce Hindi words and sentences.

During the testing process, I also checked whether the real-time text highlighting feature was properly synchronized with the Hindi speech playback. The results showed that both speech and highlighting were working correctly together.

Date – 6 February

On the sixth day, I performed browser compatibility testing to ensure that the application worked consistently across different web browsers. The testing was conducted on Google Chrome, Mozilla Firefox, and Microsoft Edge.

During this testing phase, I observed how each browser handled speech synthesis and whether the Hindi voice was available and functioning properly. The results indicated that the Hindi voice worked correctly in browsers that supported system-installed speech voices. This testing helped confirm that the application behaved consistently across different environments.

Date – 7 February

On the final day of the week, I documented the entire troubleshooting process and the steps taken to resolve the Hindi speech synthesis issue. This documentation included the problem identification, investigation process, system configuration steps, and the testing results obtained during the week.

Creating this documentation was important because it provides a clear explanation of the issue and the solution. It will also help other developers or team members understand how to enable Hindi speech synthesis support in similar applications.